

THE COMPLETE ENUMERATION OF STRICTLY DEZA GRAPHS WITH AT MOST 21 VERTICES

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Based on joint project with S. Goryainov and L. Shalaginov

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Strongly regular graphs

A k -regular graph on v vertices is called a **strongly regular graph** with parameters (v, k, λ, μ) , if any two adjacent vertices in this graph have λ common neighbours and any two distinct non-adjacent vertices have μ common neighbours.

Deza graphs

Deza graphs were introduced in 1999 [1] as a generalisation of strongly regular graphs.

A k -regular graph on v vertices is called a **Deza graph** with parameters (v, k, b, a) , $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b .

A Deza graph is called a **strictly Deza graph** if it has diameter 2 and is not strongly regular.

[1] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Parameters of Deza graphs

Let N_v be the neighbourhood of vertex v : $N_v = \{u : u \sim v\}$.

For fixed vertex u of Deza graph Γ we define

$$\alpha = |\{v \in \Gamma : |N_u \cap N_v| = a\}|,$$

$$\beta = |\{v \in \Gamma : |N_u \cap N_v| = b\}|.$$

Lemma 1 ([1, Proposition 1.1])

α and β do not depend on u and

$$\alpha = \begin{cases} \frac{b(n-1) - k(k-1)}{b-a}, & \text{if } a \neq b, \\ \frac{k(k-1)}{a}, & \text{if } a = b; \end{cases}$$

$$\beta = \begin{cases} \frac{a(n-1) - k(k-1)}{a-b}, & \text{if } a \neq b, \\ \frac{k(k-1)}{a}, & \text{if } a = b. \end{cases}$$

Parameters of Deza graphs

Lemma 2 ([1, Corollary 1.2])

Let Γ be Deza graph with parameters $(v, k, b, a), \beta, \alpha$. Then

- (1) $b - a$ divides $b(n - 1) - k(k - 1)$;
- (2) if $\alpha \neq 0$, then $n \geq 2k - a$;
- (3) if $\alpha, \beta \neq 0$, then $a(n - 1) < k(k - 1) < b(n - 1)$.

Enumeration of strictly Deza graphs

In 1999 the complete list of strictly Deza graphs with at most 13 vertices was presented (see [1]).

In 2011 an algorithm for enumeration of Deza graphs was presented by S. Goryainov and L. Shalaginov and this list was extended up to 16 vertices (see [2]).

[2] S. V. Goryainov, L. V. Shalaginov, On Deza graphs with 14, 15, and 16 vertices, *Sib. Electron. Mat. Izv.*, **8**, (2011) 105–115

Original algorithm: preparations

The first step is to obtain the list of possible parameters of Deza graphs with the restrictions from Lemma 2.

Then the algorithm produces adjacency matrices of Deza graphs with required parameters, where each row represented as single integer number in binary form.

To accomplish that we construct three special arrays before we start the enumeration of matrices:

NumberOfOnes, one-dimensional array which contains the number of 1s in binary form of number n for each n ;

NumberOfCombinations, two-dimensional array which contains the number of combinations $\binom{n}{k}$ for each n and k ;

Combinations, three-dimensional array which contains all possible combinations of k 1s in n -digit binary number for each n and k .

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10-----	<----- Combinations[6][3] =	111000	011100
1-0-----		110100	011010
1--0----		110010	011001
1---0---		110001	010110
0----0--		101100	010101
0-----0-		101010	010011
0-----0		101001	001110
		100110	001101
		100101	001011
		100011	000111

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10111000 <---- Combinations[6][3] = 111000 ! 011100

110----- 110100 011010

11-0---- 110010 011001

11--0--- 110001 010110

00---0-- 101100 010101

00----0- 101010 010011

00-----0 101001 001110

100110 001101

100101 001011

100011 000111

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100	<----- Combinations[6][3] =	111000 X	011100
110-----		110100 !	011010
11-0----		110010	011001
10--0---		110001	010110
01---0--		101100	010101
00----0-		101010	010011
00-----0		101001	001110
		100110	001101
		100101	001011
		100011	000111

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

110----- <----- Combinations[5][2] = 11000 01010

11-0---- 10100 01001

10--0--- 10010 00110

01---0-- 10001 00101

00----0- 01100 00011

00-----0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010 <---- Combinations[5][2] = 11000 X 01010

1110---- 10100 X 01001

100-0--- 10010 ! 00110

010--0-- 10001 00101

001---0- 01100 00011

000----0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010

1110---- <---- Combinations[4][1] = 1000 0010

100-0--- 0100 0001

010--0--

001---0-

000----0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010

11100001 <---- Combinations[4][1] = 1000 X 0010 X

10000--- 0100 X 0001 !

0100-0--

0010--0-

0001---0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010

11100001

10000--- <---- Combinations[3][3] = 111

0100-0--

0010--0-

0001---0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010

11100001

10000--- <---- Combinations[3][3] = 111 X

0100-0--

0010--0-

0001---0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010010

1110---- <---- Combinations[4][1] = 1000 X 0010 X

100-0--- 0100 X 0001 X

010--0--

001---0-

000----0

Original algorithm: example

parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000

10110100

11010001 <---- Combinations[5][2] = 11000 X 01010

1110---- 10100 X 01001

100-0--- 10010 X 00110

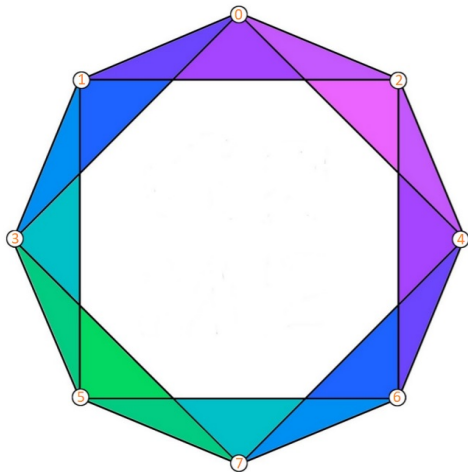
010--0-- 10001 ! 00101

000---0- 01100 00011

001----0

Original algorithm: example

01111000
10110100
11001010
11000101
10100011
01010011
00101101
00011110



Improvements: the first two rows

Let us consider three possible cases:

1. $\alpha < k$. Then we can find two adjacent vertices which have exactly b common neighbors. In that case the first two rows of the adjacency matrix look like this:

$$\begin{array}{cccc} & \overbrace{1 \dots 1}^b & \overbrace{1 \dots 1}^{k-b-1} & \overbrace{0 \dots 0}^{k-b-1} & \overbrace{0 \dots 0}^{n-2k+b} \\ 01 & & & & \\ 10 & 1 \dots 1 & 0 \dots 0 & 1 \dots 1 & 0 \dots 0 \end{array}$$

2. $\alpha = k$. Then we can find two adjacent vertices which have exactly b common neighbors, because we obtain a strongly regular graph when we can't find such vertices. In that case the first two rows of the adjacency matrix look the same as in case 1.

Improvements: the first two rows

3. $\alpha > k$. Then we can find two non-adjacent vertices which have exactly a common neighbors. In that case the first two rows of the adjacency matrix look like this:

$$\begin{array}{cccc} & \overbrace{1 \dots 1}^a & \overbrace{1 \dots 1}^{k-a} & \overbrace{0 \dots 0}^{k-a} & \overbrace{0 \dots 0}^{n-2k+a-2} \\ 00 & 1 \dots 1 & 1 \dots 1 & 0 \dots 0 & 0 \dots 0 \\ 00 & 1 \dots 1 & 0 \dots 0 & 1 \dots 1 & 0 \dots 0 \end{array}$$

Improvements: the next rows

Assume we have the first two rows of the adjacency matrix and trying to construct the third row:

```
011  11111100000
101  11100011100
110  -----
```

Notice that columns can be divided into four blocks

```
011  111|111|000|00
101  111|000|111|00
110  ---|---|---|--
```

and changing the order of 1s inside each block gives an equivalent matrix.

Improvements: the next rows

For example these two matrices are equivalent:

011	111 111 000 00	011	111 111 000 00
101	111 000 111 00	101	111 000 111 00
110	110 100 110 10	110	101 001 011 10

For the next rows, this division into blocks can be extended, but with each row the number of blocks multiplies by 2 (8 blocks for the 4th row, 16 blocks for the 5th row, etc.).

- We use Magma to check whether graphs are isomorphic after the completion of the enumeration. We also use Magma after adding the 3rd, the 4th and the 5th row.
- In case of $a = 0$ we use Magma to calculate the diameter of the resulting graphs. If the diameter does not equal 2, then this graph is not strictly Deza graph.

Table of parameters

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(8, 4, 2, 0)	1	1	0	0
(8, 4, 2, 1)	1	1	1	1
(8, 5, 4, 2)	1	1	1	1
(9, 4, 2, 1)	2	1	1	1
(10, 5, 4, 2)	1	1	1	1
(12, 5, 2, 1)	1	1	1	1
(12, 6, 3, 2)	2	1	1	1
(12, 7, 4, 3)	1	1	1	1
(12, 7, 6, 2)	1	1	1	1
(12, 9, 8, 6)	1	1	1	1
(13, 8, 5, 4)	1	1	1	1
(14, 9, 6, 4)	1	1	1	1
(15, 6, 3, 1)	1	0	0	0
(16, 5, 2, 1)	1	1	1	0
(16, 7, 4, 2)	2	2	2	2
(16, 8, 4, 2)	1	1	1	1
(16, 9, 6, 4)	2	1	1	1
(16, 9, 8, 2)	1	1	1	1
(16, 11, 8, 6)	1	1	1	1
(16, 12, 10, 8)	1	1	1	1
(16, 13, 12, 10)	1	1	1	1
(17, 8, 4, 3)	3	0	0	0

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(18, 8, 4, 2)	7	2	2	1
(18, 8, 4, 3)	1	0	0	0
(18, 9, 6, 4)	2	1	1	1
(18, 9, 8, 4)	2	1	1	1
(18, 13, 12, 8)	1	1	1	1
(19, 6, 2, 1)	6	1	1	1
(19, 8, 4, 2)	3	0	0	0
(19, 12, 8, 7)	2	0	0	0
(20, 6, 2, 1)	37	1	0	1
(20, 7, 3, 2)	1	1	1	1
(20, 7, 6, 2)	2	1	1	1
(20, 8, 4, 2)	5	1	1	0
(20, 10, 6, 4)	6	1	1	1
(20, 11, 10, 2)	1	1	1	1
(20, 13, 9, 8)	1	1	1	1
(20, 13, 12, 8)	1	1	1	1
(20, 14, 10, 9)	1	1	0	1
(20, 17, 16, 14)	1	1	1	1
(21, 8, 3, 2)	3	1	1	1
(21, 8, 4, 2)	19	1	1	1
(21, 10, 5, 4)	4	0	0	0
(21, 10, 6, 3)	1	0	0	0
(21, 12, 7, 5)	2	1	1	1
(21, 12, 7, 6)	2	2	2	2

Cayley graphs

Let G be a group and S be a set with the property: $S = S^{-1}$, that is $\forall s \in S \exists s^{-1} \in S$ and $e \notin S$.

Cayley graph $\text{Cay}(G, S)$ of the group G with the generating set S is a graph whose vertices are elements of the group G and the element g is connected by an edge exactly to those elements that are obtained by multiplying g by elements from S .

Symmetric association schemes

Let X be a set of size n , and $R_0, R_1, \dots, R_d$ be relations defined on X . Let $A_0, A_1, \dots, A_d$ be the 0-1 matrices corresponding to the relations, that is, the (x, y) -entry of A_i is 1 if and only if $(x, y) \in R_i$. Then $(X, \{R_i\}_{i=0}^d)$ is called a **d-class symmetric association scheme** if

- (1) $A_0 = I$;
- (2) $\sum A_i = J$;
- (3) each A_i is symmetric;
- (4) for each pair i and j , $A_i A_j = \sum_k p_{ij}^k A_k$ for some constants p_{ij}^k .

Deza graphs from symmetric association schemes

Let $(X, \{R_0, R_1, \dots, R_d\})$ be a symmetric association scheme, and $F \subset \{1, 2, \dots, d\}$. Let G be the graph with adjacency matrix $\sum_{f \in F} A_f$. Then G can be a Deza graph.

Table of parameters

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(8, 4, 2, 0)	1	1	0	1
(8, 4, 2, 1)	1	1	1	1
(8, 5, 4, 2)	1	1	1	1
(9, 4, 2, 1)	2	1	1	1
(10, 5, 4, 2)	1	1	1	1
(12, 5, 2, 1)	1	1	1	1
(12, 6, 3, 2)	2	1	1	1
(12, 7, 4, 3)	1	1	1	1
(12, 7, 6, 2)	1	1	1	1
(12, 9, 8, 6)	1	1	1	1
(13, 8, 5, 4)	1	1	1	1
(14, 9, 6, 4)	1	1	1	1
(15, 6, 3, 1)	1	0	0	0
(16, 5, 2, 1)	1	1	1	0
(16, 7, 4, 2)	2	2	2	2
(16, 8, 4, 2)	1	1	1	1
(16, 9, 6, 4)	2	1	1	1
(16, 9, 8, 2)	1	1	1	1
(16, 11, 8, 6)	1	1	1	1
(16, 12, 10, 8)	1	1	1	1
(16, 13, 12, 10)	1	1	1	1
(17, 8, 4, 3)	3	0	0	0

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(18, 8, 4, 2)	7	2	2	1
(18, 8, 4, 3)	1	0	0	0
(18, 9, 6, 4)	2	1	1	1
(18, 9, 8, 4)	2	1	1	1
(18, 13, 12, 8)	1	1	1	1
(19, 6, 2, 1)	6	1	1	1
(19, 8, 4, 2)	3	0	0	0
(19, 12, 8, 7)	2	0	0	0
(20, 6, 2, 1)	37	1	0	1
(20, 7, 3, 2)	1	1	1	1
(20, 7, 6, 2)	2	1	1	1
(20, 8, 4, 2)	5	1	1	0
(20, 10, 6, 4)	6	1	1	1
(20, 11, 10, 2)	1	1	1	1
(20, 13, 9, 8)	1	1	1	1
(20, 13, 12, 8)	1	1	1	1
(20, 14, 10, 9)	1	1	0	1
(20, 17, 16, 14)	1	1	1	1
(21, 8, 3, 2)	3	1	1	1
(21, 8, 4, 2)	19	1	1	1
(21, 10, 5, 4)	4	0	0	0
(21, 10, 6, 3)	1	0	0	0
(21, 12, 7, 5)	2	1	1	1
(21, 12, 7, 6)	2	2	2	2

Table of parameters

Demo web-page with a table of parameters is available at:
<http://alg.imm.uran.ru/dezagraphs/dezatab.html>

Thanks for your attention!